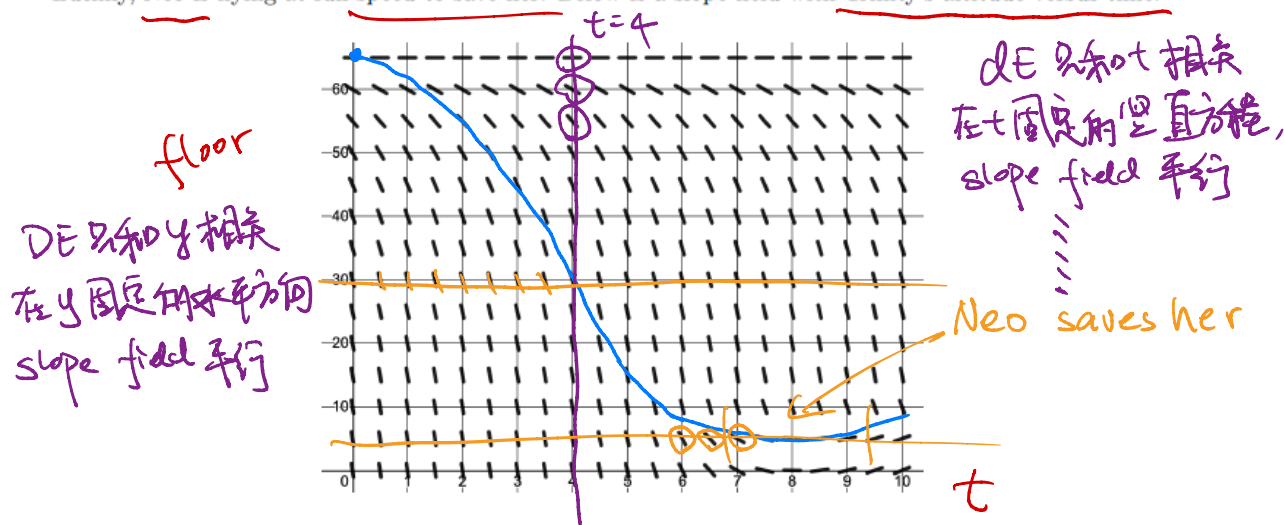


- 14 In the movie "The Matrix Reloaded", Trinity is fighting an Agent and jumps out of the sixty fifth floor. (10 marks)

Luckily, Neo is flying at full speed to save her. Below is a slope field with Trinity's altitude versus time.



Each unit in the vertical axis y corresponds to 1 floor and the units in the horizontal axis t are seconds.

- (a) (2 marks) In the movie, Neo manages to save Trinity. In the slope field above, sketch a possible trajectory for Trinity where she does not hit the ground. Make sure your sketch is clearly visible.
- (b) (2 marks) At what time does Neo arrive to save Trinity? Which features in the slope field allow you to identify this?

At $t = 8$ s, Neo arrives to save Trinity. At $t = 8$ s, the slope field is horizontal, meaning the speed of Trinity is approaching zero, so Neo must have saved her.

Time $\approx$ seconds

(c) (3 marks) This slope field is generated by a differential equation. Which of the following types of differential equations could give this slope field? (Select all options that apply)

- ☐ $\frac{dy}{dt} = f(t)$ $\rightarrow$ 只和 t 相关
- ☐ $\frac{dy}{dt} = g(y)$ $\rightarrow$ 只和 y 相关
- ☒ $\frac{dy}{dt} = h(t, y)$ $\rightarrow$ 和 t, y 都相关

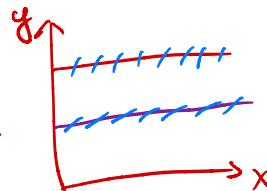
Justify your answer below.

The slope field doesn't stay constant as we move from left to right, or up to down. This means both y and t affect the value of the slope.

Q2. Consider the differential equation

$$\frac{dy}{dx} = y^2$$

只和y相关



which of the following functions is a solution for this differential equation?

(1) $y(x) = \frac{1}{a-x}$ $\int \frac{1}{y} dy = \ln|y|$

(2) $y(x) = e^x$

(3) $y(x) = \frac{x}{2-x}$

(4) $y(x) = x$

(5) $y(x) = \sin x$

$$\int \frac{1}{y^2} dy = \int dx$$

$$y^{-1} \cdot (-1) = x + C$$

$$\int y^2 dy = \int dx$$

$$y^{-1} = -x - C$$

$$\frac{1}{y} = -x + C$$

$$\frac{y^{-2+1}}{-2+1} = x + C \quad \therefore y = \frac{1}{-x+C}$$

Q3. Consider the slope field of the differential equation

$$\frac{dy}{dx} = y^3 - x^2$$

常数

Along which of the following curves are the slopes constant?

(1) $y = x + 1$

(2) $y^2 = x^3 + 2$

(3) $y^3 = x^2 + 3$ $y^3 - x^2 = 3$ constant

(4) $3y^2 = 2x + 4$

slope是常数, 可以是0, -1, +3, -5...

slope是0 $\rightarrow$ eg solution

$$\frac{dy}{dx} = C \rightarrow y^3 - x^2 = C$$

Q6. Is it true that the integral

$$\int_1^{\infty} \frac{5 \sin(t^3 + 1)}{t^2} dt$$

converges?

$$\because -1 \leq \sin(t^3 + 1) \leq 1$$

$$-\frac{5}{t^2} \leq \frac{5 \sin(t^3 + 1)}{t^2} \leq \frac{5}{t^2}$$

对称性: $f(x), g(x) > 0$ $\because -\frac{5}{t^2} < 0 \rightarrow$ 我们的目标方程
不能这样 $-\frac{5}{t^2}$

$$\int_1^{\infty} \frac{5}{t^2} dx \quad \because p=2 > 1$$

$\therefore$ converge

By comparison,

smaller integral $\int_1^{\infty} \frac{5 \sin(t^3 + 1)}{t^2} dx$ converges

Which of the following are improper integrals? Explain why.

- ☐ (a) $\int_1^{\infty} \frac{\sin x}{x} dx$ Yes
- ☐ (b) $\int_4^5 \frac{1}{x} dx$ $x \neq 0 \Rightarrow$ No
- ☐ (c) $\int_3^4 \frac{1}{\sin x} dx$ $\sin x \neq 0$ $\Rightarrow$ Yes
- ☐ (d) $\int_{-3}^3 x^{-1/3} dx$ $\frac{1}{\sqrt[3]{x}}$ $x \neq 0 \Rightarrow$ Yes
- ☐ (e) $\int_{-10}^{10} f(x) dx$ where $f(x) = \begin{cases} \frac{1}{x} + 2, & -10 \leq x < -1 \\ \frac{1}{x+2}, & -1 \leq x \leq 10 \end{cases}$ $x \neq 0$
 $\hookrightarrow$ continuous $x \neq -2$ continuous
- $\therefore$ not an improper.

Calculate $\int_0^{\infty} x e^{-4x^2} dx$ if it converges. You may calculate the limit by appealing to the dominance of one function over another, or by l'Hopital's rule.

NOTE: Enter the exact answer or round to three decimal places.

If the integral does not converge, indicate that using the checkbox.

$\int_0^{\infty} x e^{-4x^2} dx =$ ☐ The integral does not converge.

$$\begin{aligned} & \int_0^{\infty} x \cdot e^{-4x^2} dx \\ &= \lim_{R \rightarrow +\infty} \int_0^R x \cdot e^{-4x^2} dx \\ &= \lim_{R \rightarrow +\infty} \left. -\frac{1}{8} e^{-4x^2} \right|_0^R \\ &= \lim_{R \rightarrow +\infty} \left(-\frac{1}{8} e^{-4R^2} - \left(-\frac{1}{8} e^0 \right) \right) \\ &= \lim_{R \rightarrow +\infty} \left(-\frac{1}{8} e^{-4R^2} + \frac{1}{8} \right) = \frac{1}{8} \end{aligned}$$

$\therefore$ integral is convergent.

7.30 (2) $\sum_{n=0}^{\infty} C_n(x-a)^n$ $\hookrightarrow$ center

3. Consider an arbitrary power series: $S(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$

We say a power series **converges** at $x = b$, if $S(b)$ is equal to a finite number. $\rightarrow$ 求和最终稳定

3.(a) $S(a)$ converges. What does it converge to?

3.(b) $S(x)$ can potentially converge at other values of x too. Either $S(x)$ converges on an interval centred at $x = a$ or it converges everywhere.

If $S(x)$ converges on an interval centred at a of length $2r$, what are the endpoints of that interval? 终点

Hint: Draw a number line and count r units in each direction away from a .

3.(c) If the radius of convergence of $T(x) = \sum_{n=0}^{\infty} c_n(x+3)^n$ is $r = 4$, what is the largest interval on which $T(x)$ converges?

3.(d) $T(x)$, from part (c), converges at which of the following: $x = -3$, $x = 3$, $x = 0$, $x = 8$, $x = 0.5$, $x = -0.5$, $x = -2$, $x = -8$, and/or $x = 10$?

a) $S(a) = C_0 + C_1 \cancel{(a-a)} + C_2 \cancel{(a-a)^2} + \dots = C_0$
 $x=a$, $S(a)$ converges to C_0

对于 any power series 来讲, 在 $x = \text{center}$ 永远收敛!

b) $\left(\overset{-R}{\leftarrow} \overset{+R}{\rightarrow} \right)_a \rightarrow \left(\overset{-r}{\leftarrow} \overset{+r}{\rightarrow} \right)_a$
 $\underbrace{\hspace{1.5cm}}_{2r}$

endpoints: $a-r, a+r$ $(a-r, a+r)$

写开区间的原因: 对于 $a-r, a+r$ 的收敛性是未知的, 我们只知道 $a-r < x < a+r$ 收敛, if 题目问 $x = a-r$ or $a+r$, power series 是否收敛? $\rightarrow$ 不确定

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c) center, a 标准形式: $\sum_{n=0}^{\infty} C_n(x-a)^n \therefore \sum_{n=0}^{\infty} C_n(x-(-3))^n$

$$\therefore a = -3$$

$$\xleftarrow{-4} \xrightarrow{+4} \\ -3$$

$$-3-4, -3+4$$

$$\therefore (-7, 1)$$

possible $[-7, 1]$ 有可能

$(-7, 1)$, or $[-7, 1)$ or $(-7, 1]$, or $[-7, 1]$ (largest $[-7, 1]$)

d) $[-7, 1]$ largest possible interval

$$x = -3 \text{ conv}$$

$$x = 3 \text{ div}$$

$$x = 10 \text{ div}$$

$$x = 0 \text{ conv}$$

$$x = 8 \text{ div}$$

$$x = 0.5 \text{ conv}$$

$$x = -0.5 \text{ con}$$

$$x = -2 \text{ conv}$$

$$x = -8 \text{ div}$$

7. Consider the predator-prey system:

$\rightarrow xy$ $+/-$

$$\begin{cases} \frac{dy}{dt} = \underbrace{y}_{\text{prey 自己增长}} - yx \\ \frac{dx}{dt} = -\underbrace{x}_{\text{predator 自己活不了}} + yx \end{cases}$$

Assume $x(0) = 1$ and $y(0) = 2$.

(a) Determine which of the variables, x or y , represents the prey population and which represents the predator population. Explain.

~~(b)~~ Use the 1st degree Taylor polynomials centred at $t = 0$ to estimate $x(1)$ and $y(1)$.

$$x(t) \approx \quad , \quad x(1) \approx$$

$$y(t) \approx \quad , \quad y(1) \approx$$

~~(c)~~ Use the 1st degree Taylor polynomials centred at $t = 1$ to estimate $x(2)$ and $y(2)$.

$$x(t) \approx \quad , \quad x(2) \approx$$

$$y(t) \approx \quad , \quad y(2) \approx$$

② $k > 0 \Rightarrow \frac{dT}{dt} < 0$ $25 - T < 0$ 把热放到冰箱里

① $k > 0 \Rightarrow \frac{dT}{dt} > 0$ 冷的放到热的环境

4 To which of the following scenarios could the equation

$$\frac{dT}{dt} = k(25 - T)$$

with solution $T = 25 - 20e^{-kt}$ correspond to?

$t \rightarrow$ time

$t = 0 \quad T = 25 - 20 \cdot e^0 = 5 \leftarrow$ initial temperature

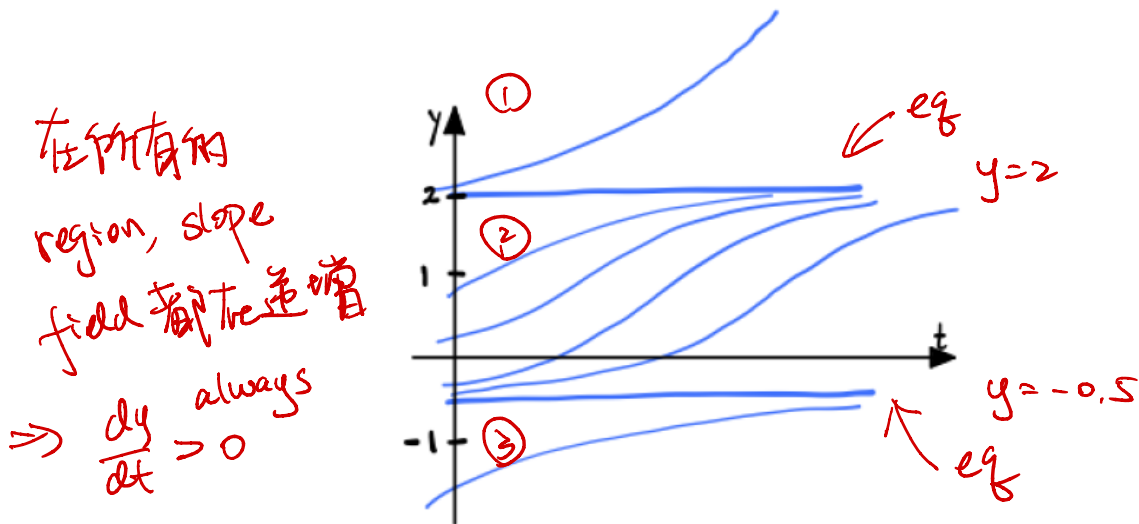
(A) A bottle of juice is bought from the store at 25 degrees and then it is placed in the fridge set at 5 degrees.

(B) A bottle of juice is taken out of a 5 degree fridge and left in a 25 degrees room.

(C) A bottle of juice is bought from the store at 20 degrees and then it is placed in the fridge set at 5 degrees.

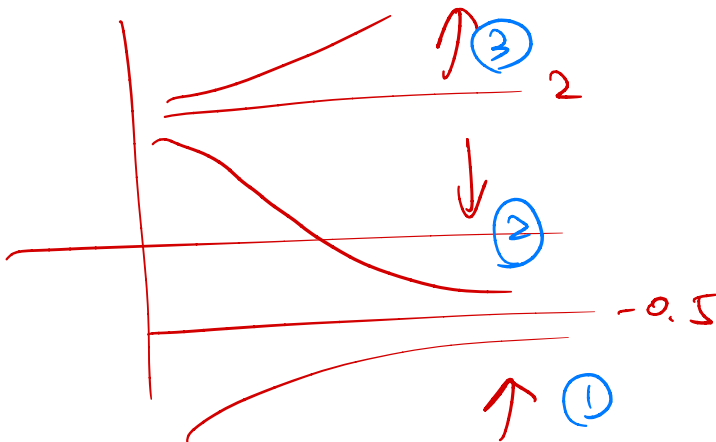
(D) A bottle of juice is taken out of a 20 degree fridge and left in a 25 degrees room.

11 Given the graph of solutions of a differential equation below, write a differential equation with such solutions.



$$\frac{dy}{dt} = (y-2)^2 \left(y+\frac{1}{2}\right)^2$$

$y = |x+1|$
 $y = x+1$



③ $+/+ \Rightarrow > 0$

② $-/- \Rightarrow > 0$

① $-/+ \Rightarrow < 0$

$\frac{dy}{dt} = (y-2) \cdot (y+\frac{1}{2})$

- 15 Newton's cooling equation for the temperature of a cup of coffee when its surroundings are at 20 degrees celsius is

$$\frac{dT}{dt} = k(20 - T)$$

$t=0, T=50^\circ\text{C}$ $t=1 \rightarrow T=40^\circ\text{C}$

Suppose that, initially, the cup is at 50 C and after one hour its temperature dropped by 10 C.

- (1) Find the equilibrium solutions of Newton's equation above and give a practical description of them.
- (2) Solve the differential equation with the conditions given in the statement above.
- (3) If we brew another cup of the same coffee to 100°C and leave it in the same room, is it possible for it to be at 80°C after one hour? Why / Why not?

1) $\frac{dT}{dt} = 0 \Rightarrow k(20 - T) = 0 \therefore T = 20^\circ\text{C}$

The temperature won't change if the coffee temperature is at 20°C (no heat transfer)

2) $T(0) = 50$ $T(1) = 40$

$$\int \frac{1}{ax+b} dx$$

$$\frac{dT}{dt} = k(20 - T) \quad \int \frac{1}{20 - T} dT = \int \frac{1}{T - 20} dT = \ln|ax+b| \cdot \frac{1}{a}$$

$$\ln|20 - T| \cdot (-1) = kt + C \quad \ln|20 - T| = -kt - C$$

$$|20 - T| = e^{-kt - C} \quad \therefore T \geq 20 \quad T - 20 = C \cdot e^{-kt}$$

$$\therefore T = 20 + C \cdot e^{-kt}$$

$$T(0) = 50 \quad 50 = 20 + C \cdot e^0 \quad \therefore C = 30$$

$$T = 20 + 30 \cdot e^{-kt}$$

$$T(1) = 40 \quad 40 = 20 + 30 \cdot e^{-k \cdot 1}$$

$$\therefore T = 20 + 30 \cdot e^{(-\ln \frac{2}{3}) \cdot t}$$

$$\begin{aligned} 30 \cdot e^{-k} &= 20 \\ \ln e^{-k} &= \ln \left(\frac{2}{3} \right) \\ -k &= \ln \left(\frac{2}{3} \right) \\ k &= -\ln \left(\frac{2}{3} \right) \end{aligned}$$

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(3) If we brew another cup of the same coffee to 100°C and leave it in the same room, is it possible for it to be at 80°C after one hour? Why / Why not?

$$T(0) = 100^\circ\text{C}$$

$$T(1) = 80^\circ\text{C}$$

$$T = 20 + 80 \cdot e^{\ln \frac{2}{3} \cdot t}$$

$$t=0, T=100$$

$$t=1$$

$$20 + 80 \cdot e^{\ln \frac{2}{3} \cdot 1} < 80^\circ\text{C}$$

$$\ln \frac{2}{3} < 0 \therefore e^{\ln \frac{2}{3}} < 1$$

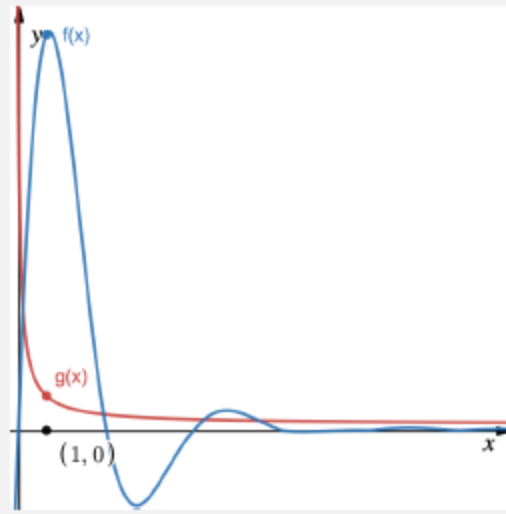
$$\therefore 20 + 80 \cdot e^{\ln \frac{2}{3}} < 80^\circ\text{C}$$

$$73^\circ\text{C}$$

after one hour, coffee temperature below 80°C

$$= 20 + 30 \cdot e^{\ln \frac{2}{3} \cdot t} //$$

Consider the functions $f(x)$ and $g(x)$.



$$x \rightarrow \infty \quad \begin{matrix} \star & \downarrow \\ g(x) & > f(x) \end{matrix}$$

- i) Suppose $\int_1^\infty f(x) dx < \infty$. What can you say about $\int_1^\infty g(x) dx$? $\rightarrow$ uncertain
- ii) Suppose $\int_1^\infty f(x) dx = \infty$. What can you say about $\int_1^\infty g(x) dx$? $\rightarrow$ diverge
- iii) Suppose $\int_1^\infty g(x) dx < \infty$. What can you say about $\int_1^\infty f(x) dx$? $\rightarrow$ converge
- iv) Suppose $\int_1^\infty g(x) dx = \infty$. What can you say about $\int_1^\infty f(x) dx$? $\rightarrow$ uncertain

Four pairs of species are given, with descriptions of how they interact along with a number of systems of ODEs. Match the pairs of species with their corresponding system of ODEs.

→ predator-prey

1. Bees/flowers: each needs the other to survive.
2. Owls/ trees: owls need trees but trees are indifferent.
3. Elk/buffalo: in competition and would do fine alone; (-/-)
4. Fox/hare: fox eats hare and needs it to survive.

- (a) $\begin{cases} \frac{dx}{dt} = -0.2x + 0.03xy \\ \frac{dy}{dt} = 0.4y - 0.08xy \end{cases}$
- (b) $\begin{cases} \frac{dx}{dt} = 0.18x \leftarrow \text{tree} \\ \frac{dy}{dt} = -0.4y + 0.3xy \leftarrow \text{owls} \end{cases}$
- (c) $\begin{cases} \frac{dx}{dt} = -0.6x + 0.18xy \\ \frac{dy}{dt} = -0.1y + 0.09xy \end{cases}$
- (d) Write a possible system of ODEs for the species pair that does not have a match.

3. $\begin{cases} \frac{dx}{dt} = +2x - 3xy \\ \frac{dy}{dt} = +2y - 3xy \end{cases}$

fine alone
competition

4. (11 marks) For each series, determine whether it is absolutely convergent, conditionally convergent or divergent. In each case, state the test you are using and justify your steps. This question continues on the next page.

(a) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$

→ This is a series (因为有 Σ), 没有 x , 不是 power

① 是否为 geometric? $\times$

② 能否用 divergence test? $\lim = 0 \times$

③ 能否用 integral test? $\checkmark$

④ 能否用 ratio test? $\times$

$$f(x) = \frac{1}{x(\ln x)^3}$$

$$\int_2^{\infty} \frac{1}{x(\ln x)^3} dx$$

$$= \lim_{R \rightarrow \infty} \int_2^R \frac{1}{x(\ln x)^3} dx$$

$$= \lim_{R \rightarrow \infty} \left[-\frac{1}{2(\ln x)^2} \right]_2^R$$

$$= \lim_{R \rightarrow \infty} \left(-\frac{1}{2(\ln R)^2} - \left(-\frac{1}{2(\ln 2)^2} \right) \right)$$

$$= +\frac{1}{2(\ln 2)^2} \rightarrow \text{converge}$$

$$\int \frac{1}{x(\ln x)^3} dx$$

$$u = \ln x \quad du = \frac{1}{x} \cdot dx$$

$$= \int \frac{1}{u^3} \cdot du = \frac{u^{-2}}{-2} + C = -\frac{1}{2} \cdot u^{-2} + C$$

$$= -\frac{1}{2} \cdot (\ln x)^{-2} + C$$

By integral test

(b) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$ converges

Ratio

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = \lim_{n \rightarrow \infty} \frac{n(n-1)(n-2) \dots 1}{n \cdot n \cdot n \cdot n \cdot n} \rightarrow 0 \rightarrow \text{divergence test } \# \nexists$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{n^n}{(n+1)^n} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n$$

$$0 \leq \lim < 1 \text{ converge} \rightarrow \therefore \sum_{n=1}^{\infty} \frac{n!}{n^n} \text{ converges!}$$

★ $\lim > 1$ div

$\lim = 1$ inconclusive

For the following series compute the radius R of convergence and the interval I of convergence. Justify your answer. Credit will only be given in so far as the reasons are adequate.

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{7^n \ln n} \rightarrow \text{power series } C_n (x-a)^n$$

$$(x-0)^n \therefore \text{center} = 0$$

power series R (ratio) ratio

$$a_{n+1} = (-1)^{n+1} \frac{x^{n+1}}{7^{n+1} \ln(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\cancel{x^{n+1}}}{\cancel{7^{n+1}} \cdot \ln(n+1)} \cdot \frac{\cancel{7^n} \ln n}{\cancel{x^n}} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{\ln n}{7 \ln(n+1)} |x|$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{7 \cdot \frac{1}{n+1}} |x| = \lim_{n \rightarrow \infty} \frac{n+1}{7n} |x| = \frac{1}{7} |x|$$

$$\therefore R = 7 \quad \text{center} = 0$$

$$\left(\frac{-7}{0}, \frac{+7}{0} \right) \Rightarrow (-7, 7) \text{ endpoints}$$

convergence are unknown

absolutely converge $\rightarrow$ 加绝对值 converge

conditional converge $\rightarrow$ 加绝对值不conv, 去掉绝对值 converge.

week 8 - week 11 必须做

复习笔试题及弄懂

2024 T12 + practice
